

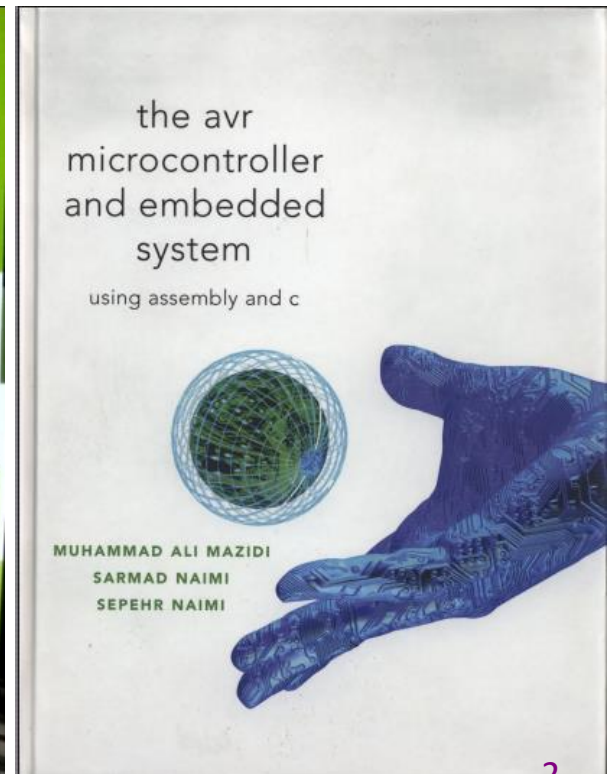
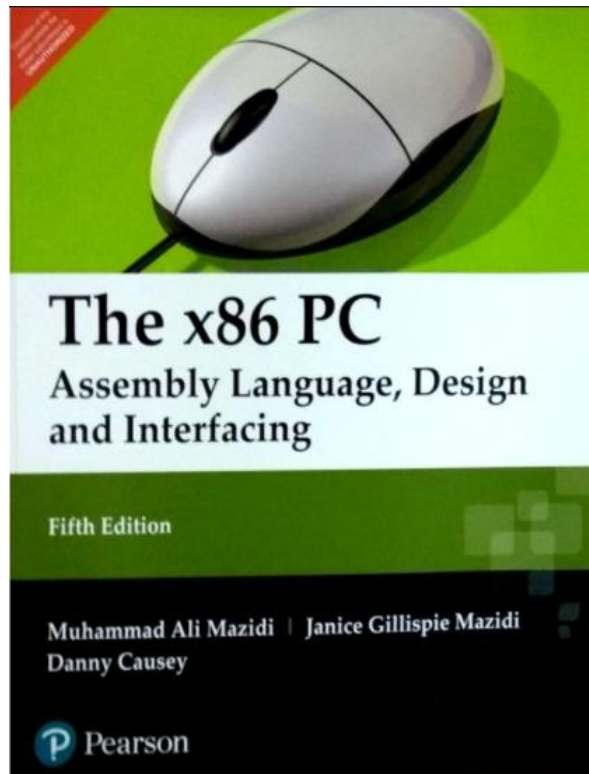
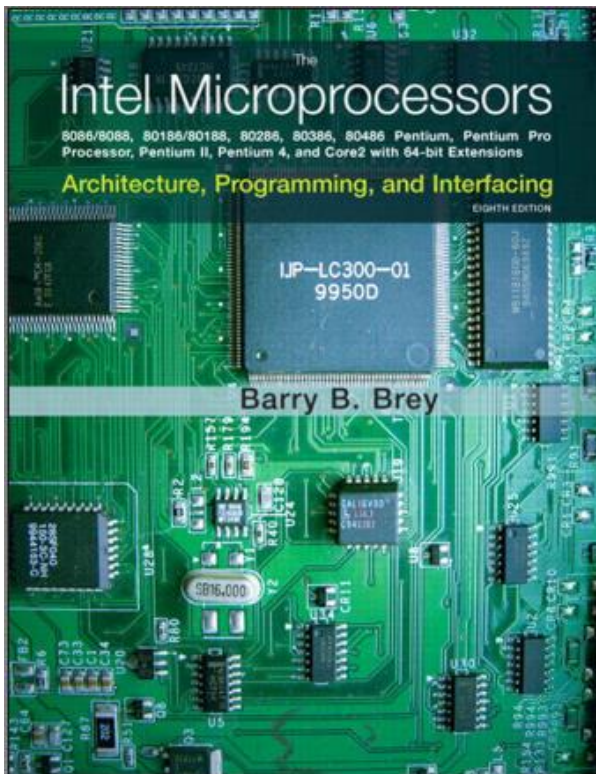
EE-303

**MICROPROCESSOR AND
INTERFACING TECHNIQUES**

Course Information

◆ Books

- ❖ “The Intel Microprocessors”, 8th Edition by Barry B. Brey, Publisher: Pearson Education, ISBN: 9780135026458
- ❖ The 80x86 PC Assembly Language, Design and Interfacing, 5th Edition by M. Ali Mazidi, Publisher: Prentice Hall Press, ISBN: 9780135026489
- ❖ The AVR Microcontroller and Embedded System Using Assembly and C (2011), by M. Ali Mazidi, Publisher: Prentice Hall Press, ISBN: 9780138003319



Course Information

◆ Class Composition



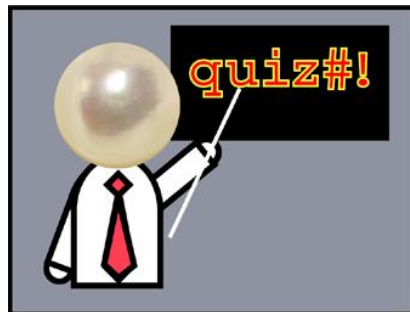
Lectures



Readings



Term Project

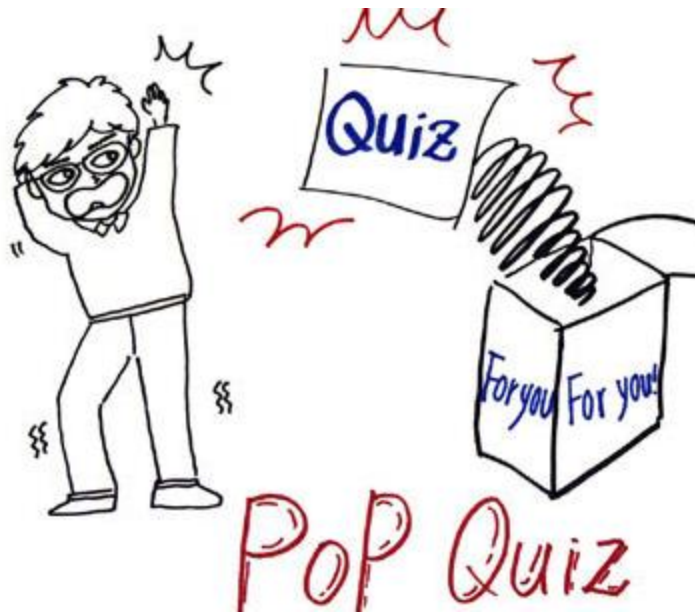


Quizzes

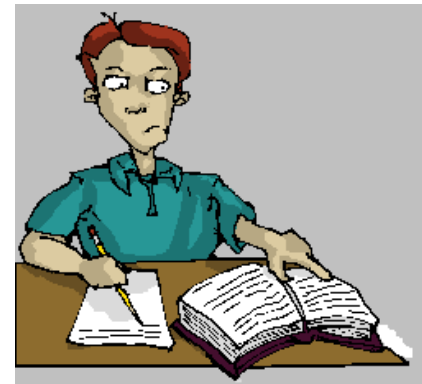
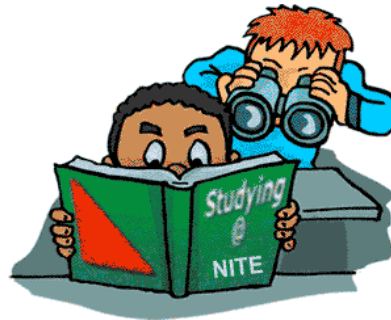


Assignments

Quizzes & Assignments



On Time



No Copying

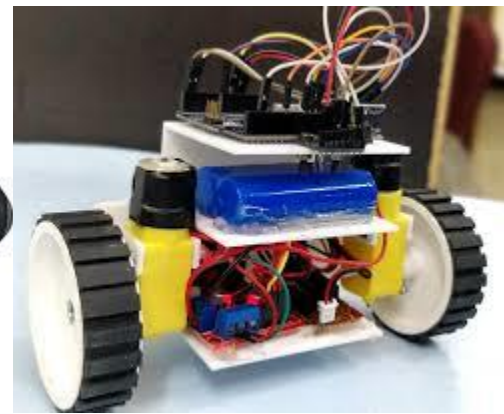
Applications of Microprocessors: General Purpose μ P

Desktop, PCs, Laptops, Dock-stations, Servers, Supercomputer



Applications of Microprocessors: Embedded Systems

Consumer Electronics: Toys, Cameras, Camcorders, Robots



Applications of Microprocessors: Embedded Systems

Consumer Products: Washing Machines, Microwaves, Ovens, Coffee Makers



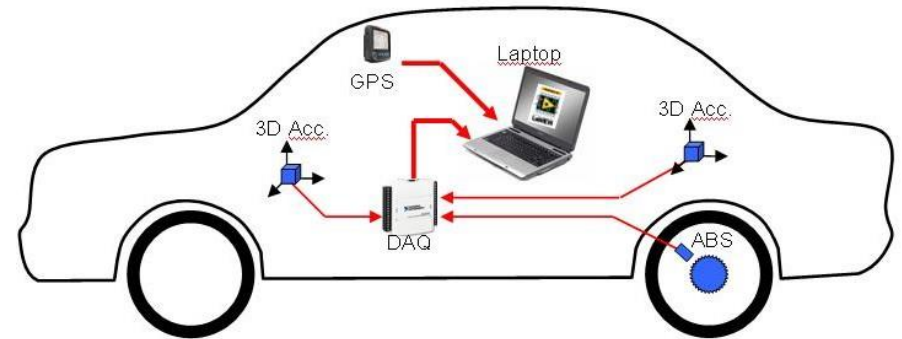
Applications of Microprocessors: Embedded Systems

Instrumentation: Digital Oscilloscope, Medical Instruments, ICUs, MRI

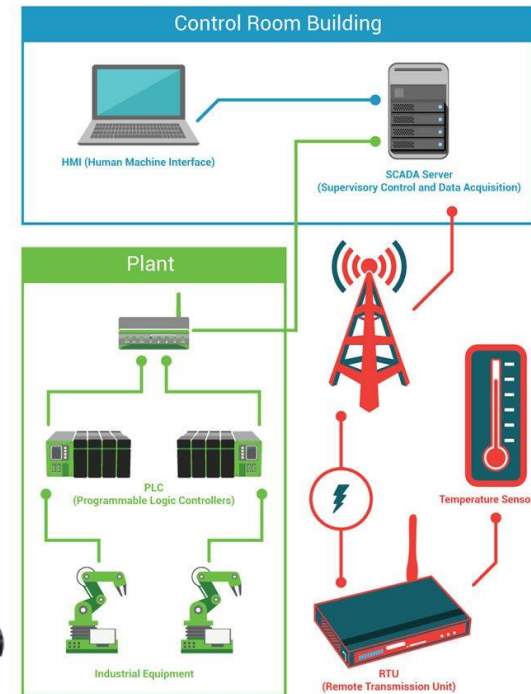
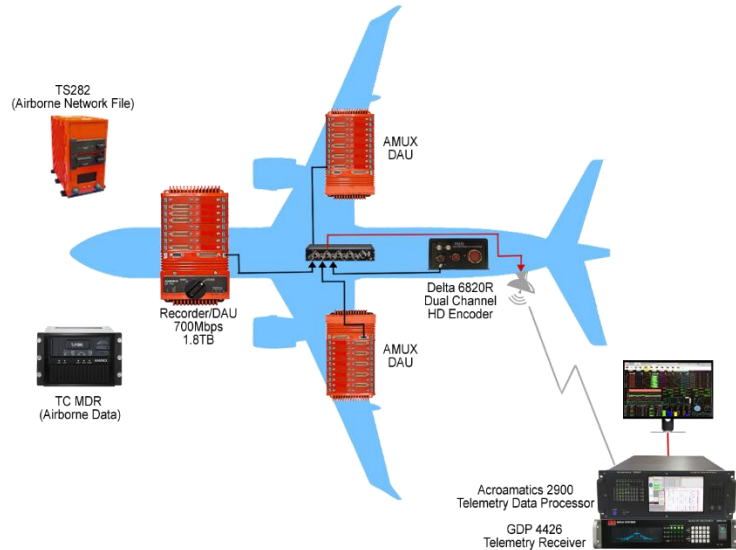


Applications of Microprocessors: Embedded Systems

Process Control: Data Acquisition, Industrial Control



Commercial Data Acquisition



Applications of Microprocessors: Embedded Systems

Communication: Telephone, Modems, Answering Phones, Printers
Cordless phones, Fax Machines

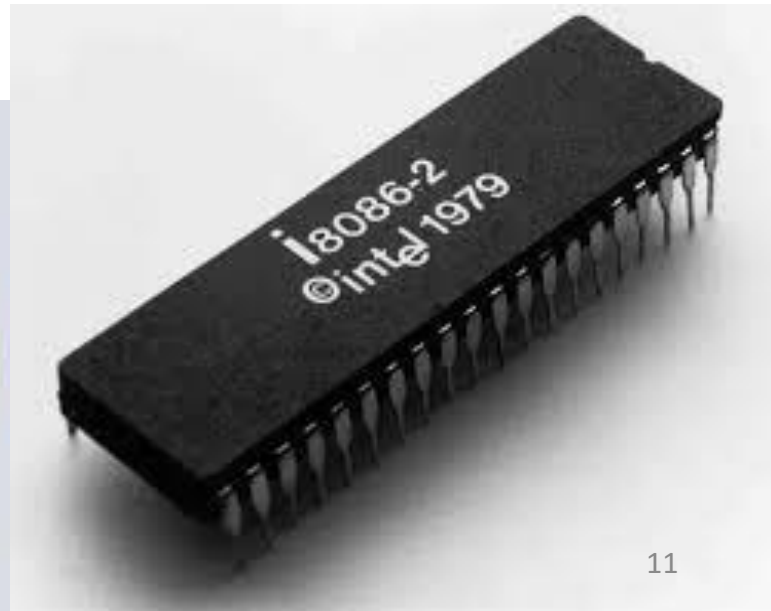
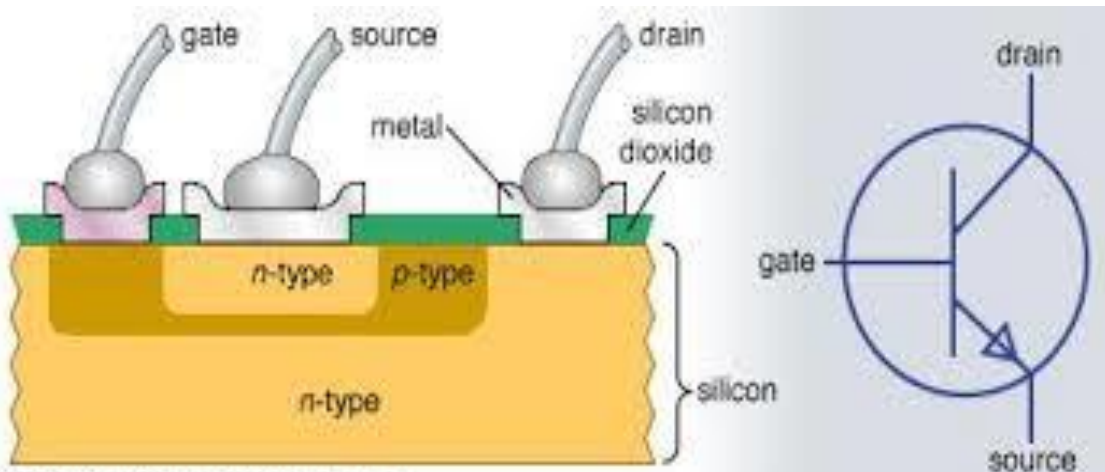


viseGEEK



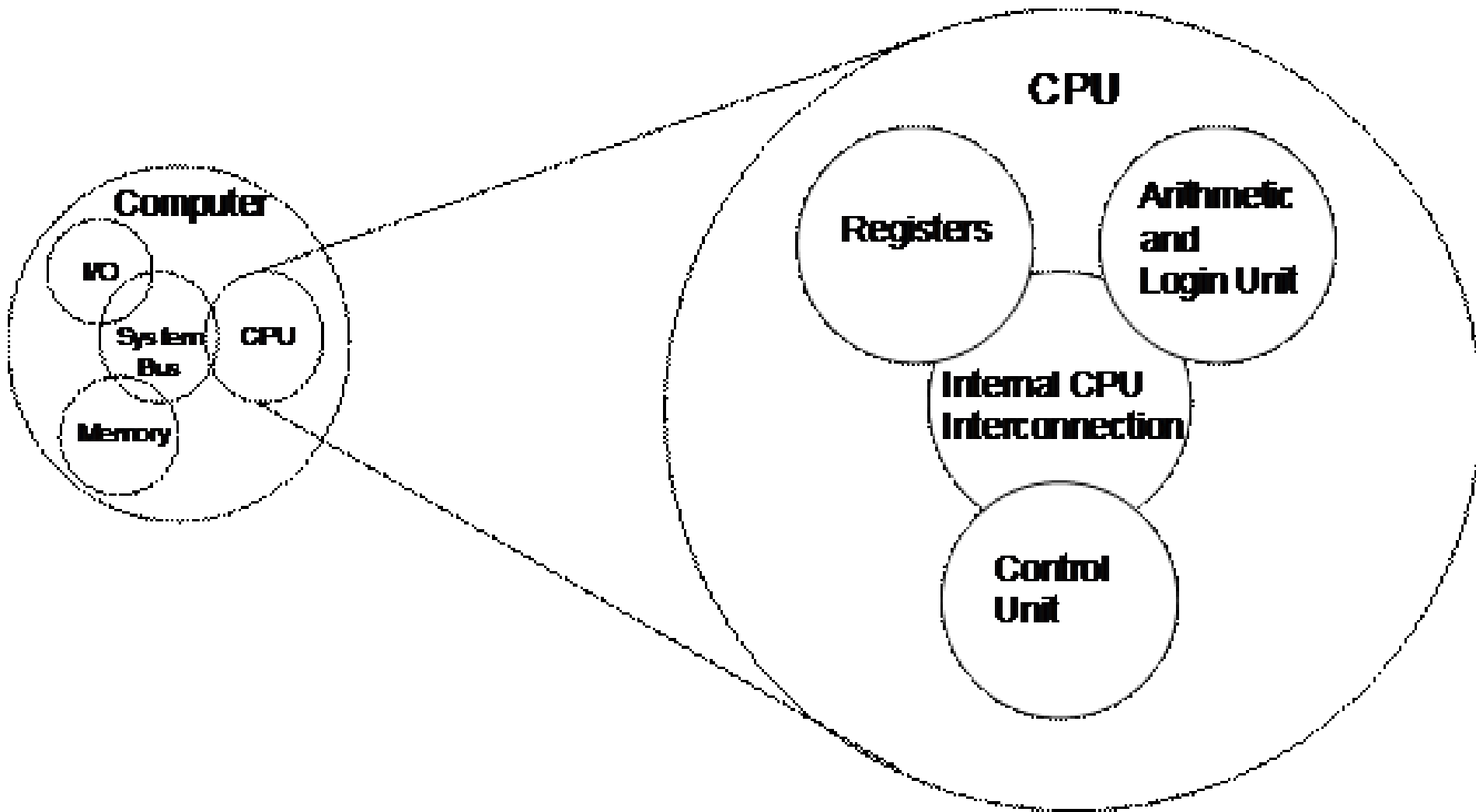
Define Microprocessor

- An integrated circuit that contains all the functions (tasks) of a computer's central processing unit.
- Microprocessor performs three major functions (tasks):
 - ✓ Data transfer (Itself, memory, I/O)
 - ✓ Arithmetic and logic Operations
 - ✓ Program flow from simple decisions



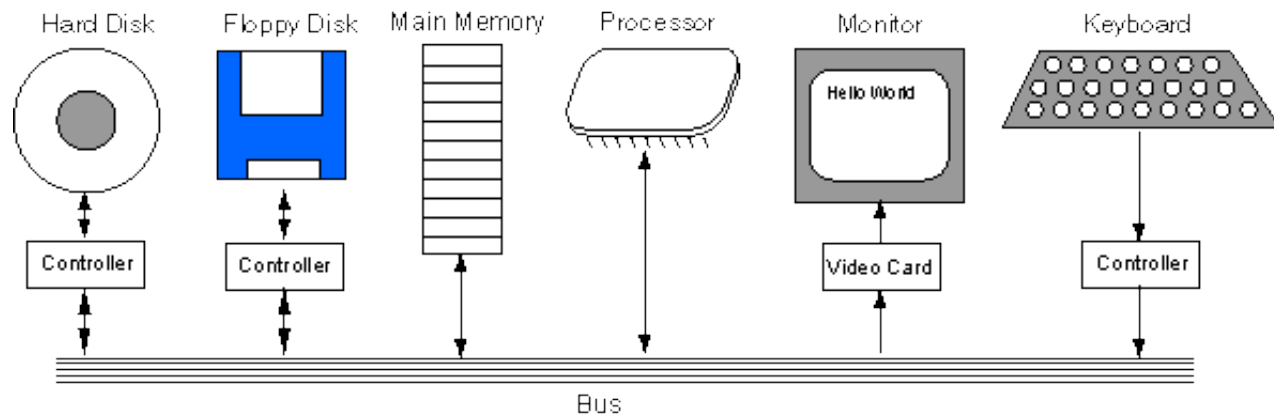
<i>Operation</i>	<i>Comment</i>	<i>Decision</i>	<i>Comment</i>
Addition		Zero	Test a number for zero or not-zero
Subtraction		Sign	Test a number for positive or negative
Multiplication		Carry	Test for a carry after addition or a borrow after subtraction
Division			
AND	Logical multiplication		
OR	Logic addition	Parity	Test a number for an even or an odd number of ones
NOT	Logical inversion		
NEG	Arithmetic inversion	Overflow	Test for an overflow that indicates an invalid result after a signed addition or a signed subtraction
Shift			
Rotate			

Functional units of a Computer and CPU



BUS

- A **bus** is set of common connections or group of wires that carry the same type of information that interconnect the sections of a computer system.
- Three types of buses exist for the transfer of information
 1. Address Bus
 2. Data Bus
 3. Control Bus



Main Components of a Computer System

Address Bus: 1-bit
Data Bus: 8-bit

	7	6	5	4	3	2	1	0
1								
0								

1H								
0H								

Address Bus: 2-bit
Data Bus: 8-bit

11								
10								
01								
00								

3H								
2H								
1H								
0H								

111
110
101
100
011
010
001
000

7H
6H
5H
4H
3H
2H
1H
0H

Address Bus: 2-bit
Data Bus: 8-bit

FH, 1111
EH, 1110
DH, 1101
CH, 1100
BH, 1011
AH, 1010
9H, 1001
8H, 1000
7H, 0111
6H, 0110
5H, 0101
4H, 0100
3H, 0011
2H, 0010
1H, 0001
0H, 0000

Address Bus: 4-bit
Data Bus: 8-bit

<u>Microprocessor</u>	<u>Data Bus Width</u>	<u>Address Bus Width</u>	<u>Memory Size</u>
8086	16	20	1M
8088	8	20	1M
80186	16	20	1M
80188	8	20	1M
80286	16	24	16M
80386 SX	16	24	16M
80386 DX	32	32	4G
80386 EX	16	26	64M
80486	32	32	4G
Pentium	64	32	4G
Pentium Overdrive	32	32	4G
Pentium Pro	64	32	4G
Pentium Pro	64	36	64G
Pentium II	64	32	4G
Pentium II	64	36	64G

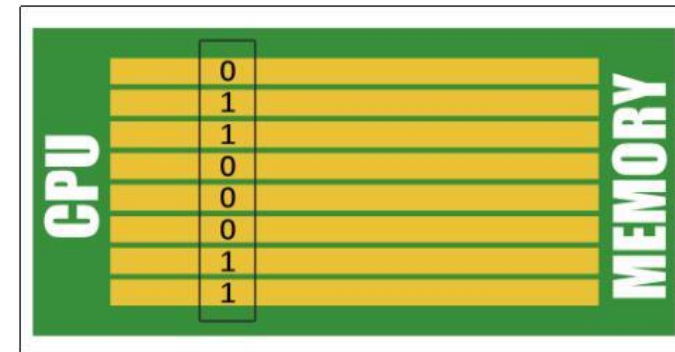
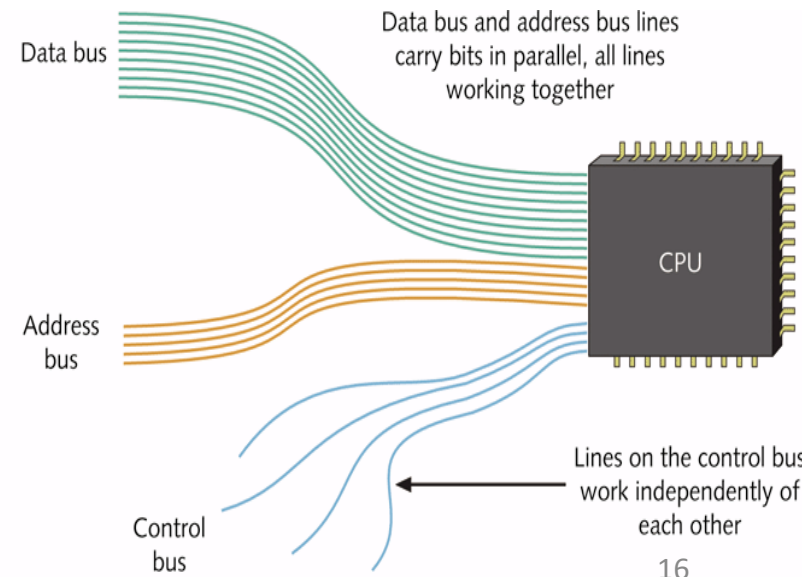
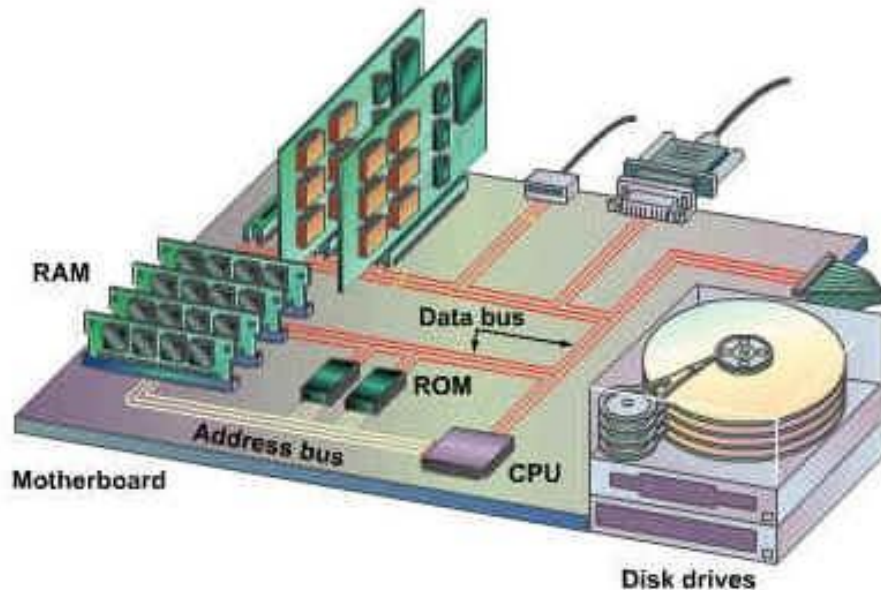
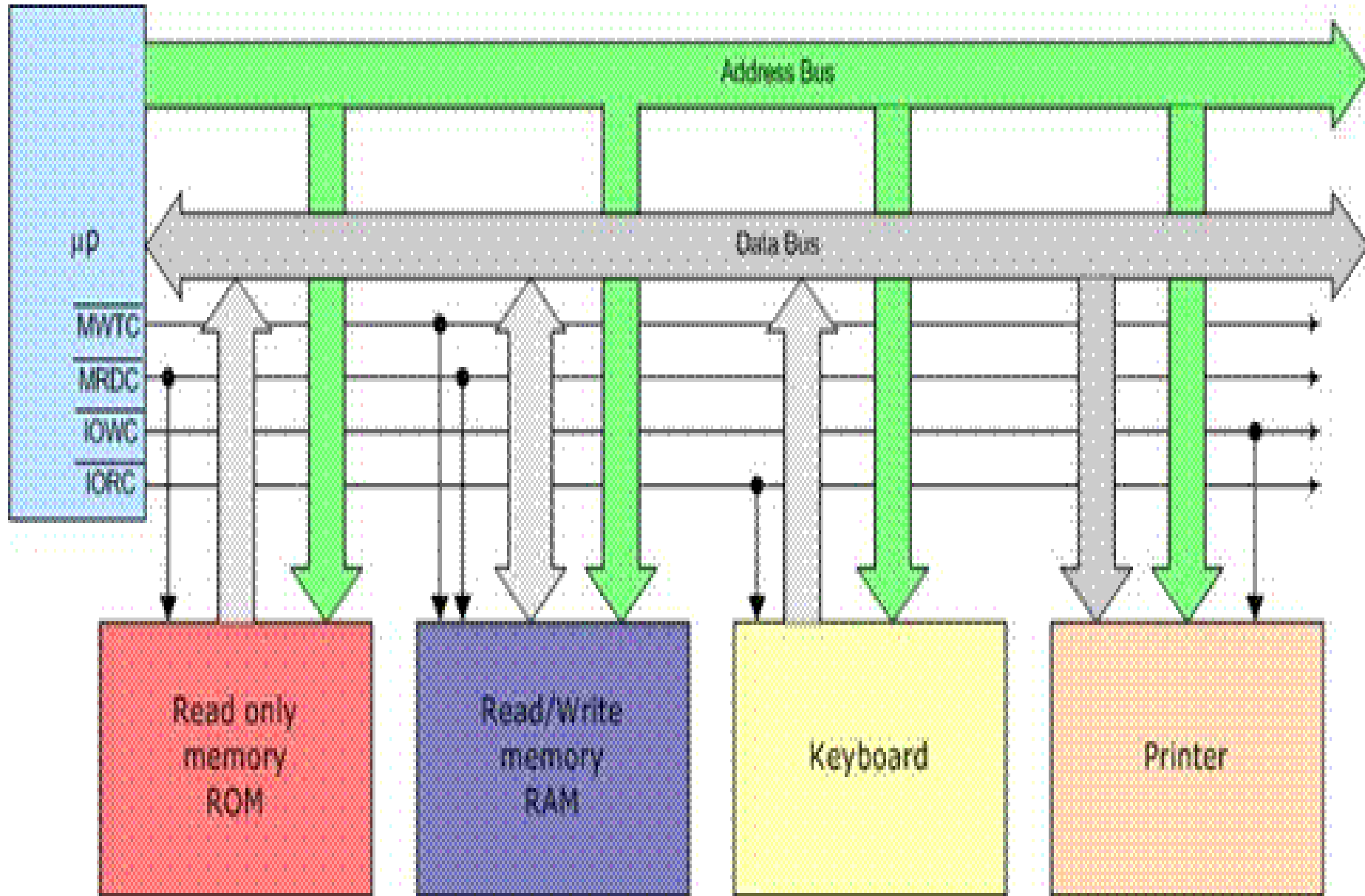


Figure 2
8 bit data bus example



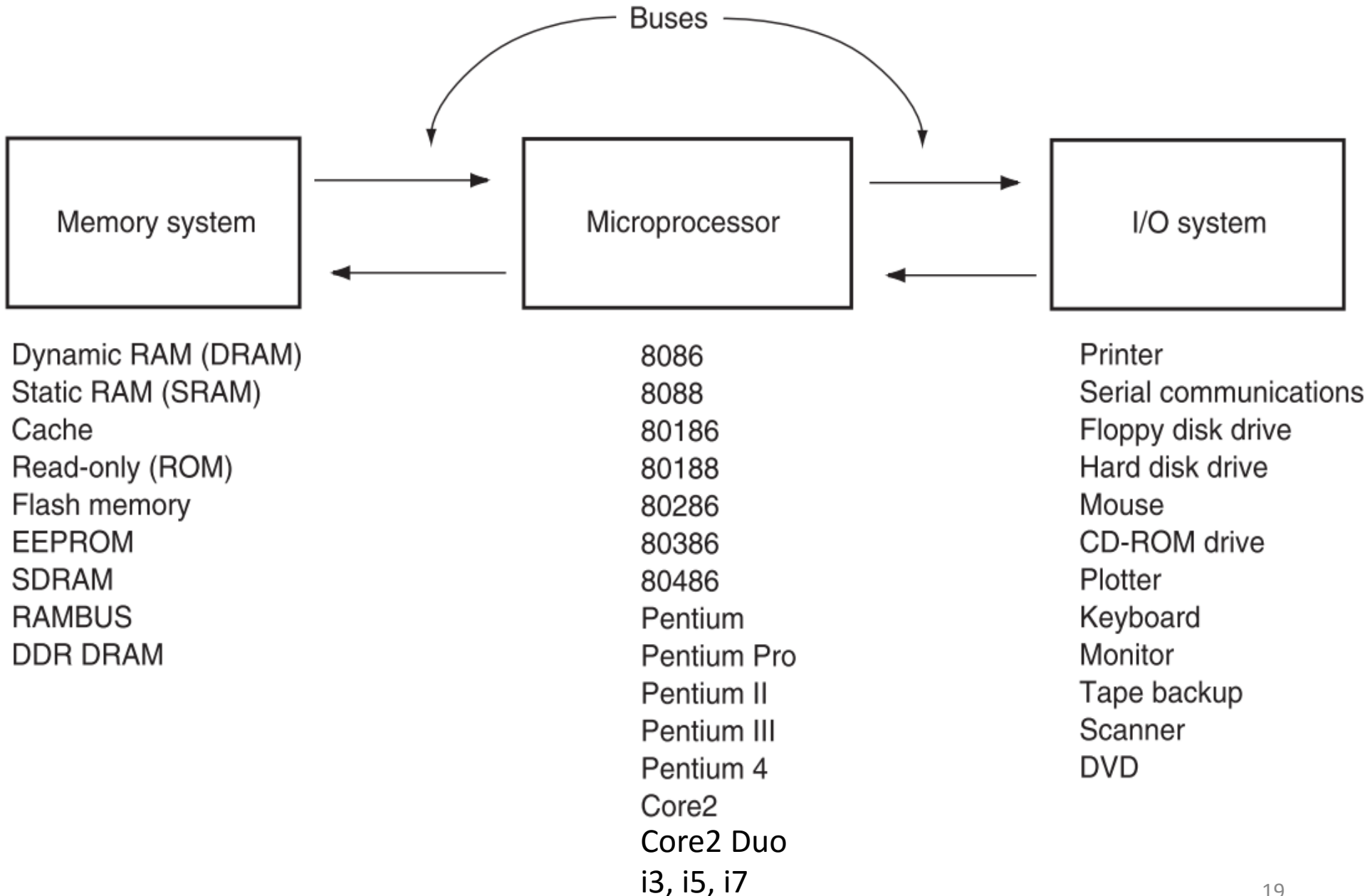
Block Diagram of Bus Structure



Bus:

- **Address bus** requests a memory location from the memory or an I/O location from the I/O devices.
 - ✓ **I/O Address bus** contains a 16-bit I/O address, from 0000H through FFFFH. The 16-bit I/O address, or port number, selects one of 64K different I/O devices.
 - ✓ **Memory address bus** contains a memory address, which varies in width with the different versions of the microprocessor.
- **Data bus** transfers information between the microprocessor and its memory and I/O address space.
 - ✓ Data transfers vary in size, from 8-bits wide to 64 bits wide in various members of microprocessors.
- **Control bus** contains lines that select the memory or I/O and cause them to perform a read or write operation. (e.g. memory read control, memory write control, I/O read control and I/O write control)

Block Diagram of a Microprocessor-Based Computer System

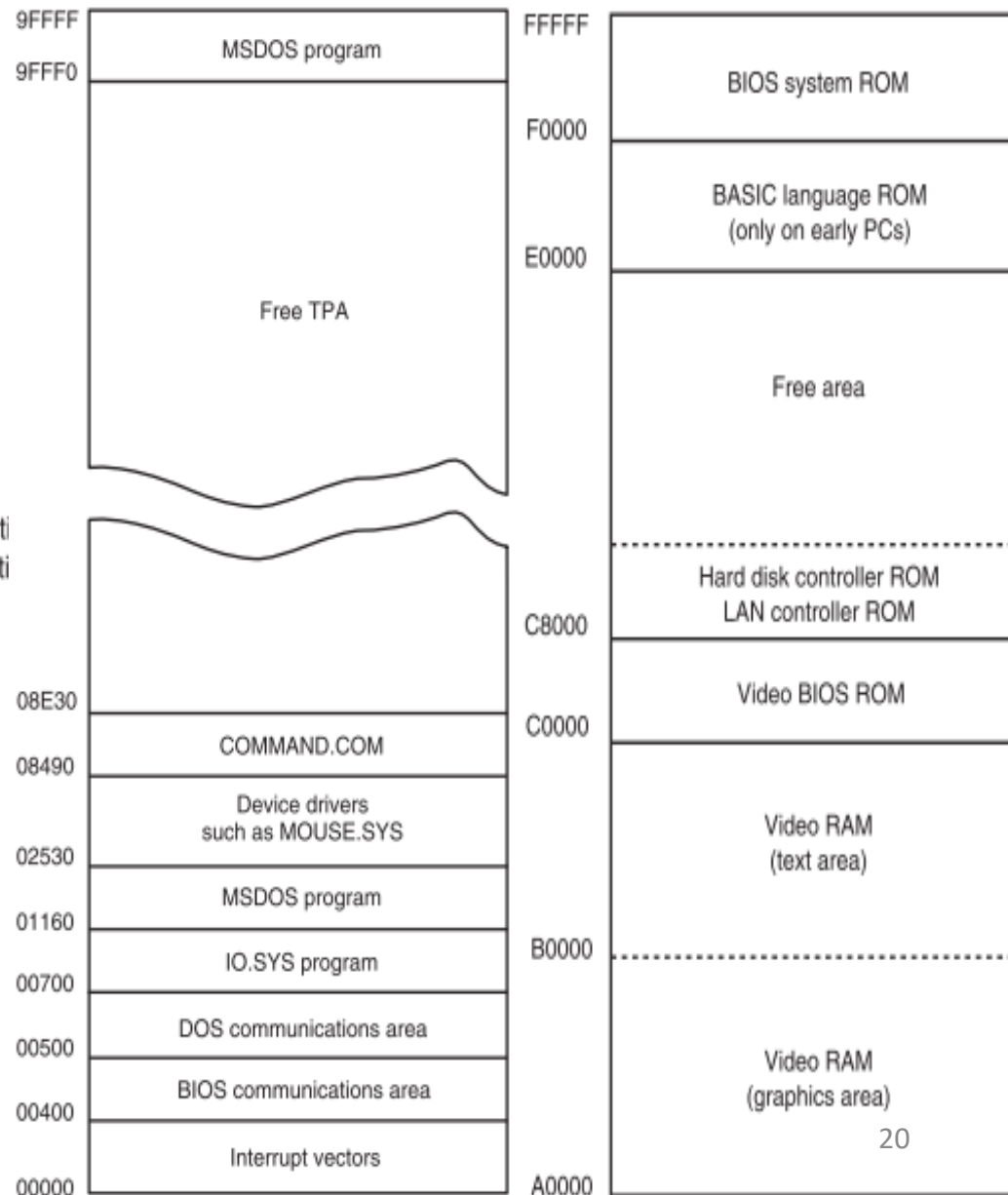
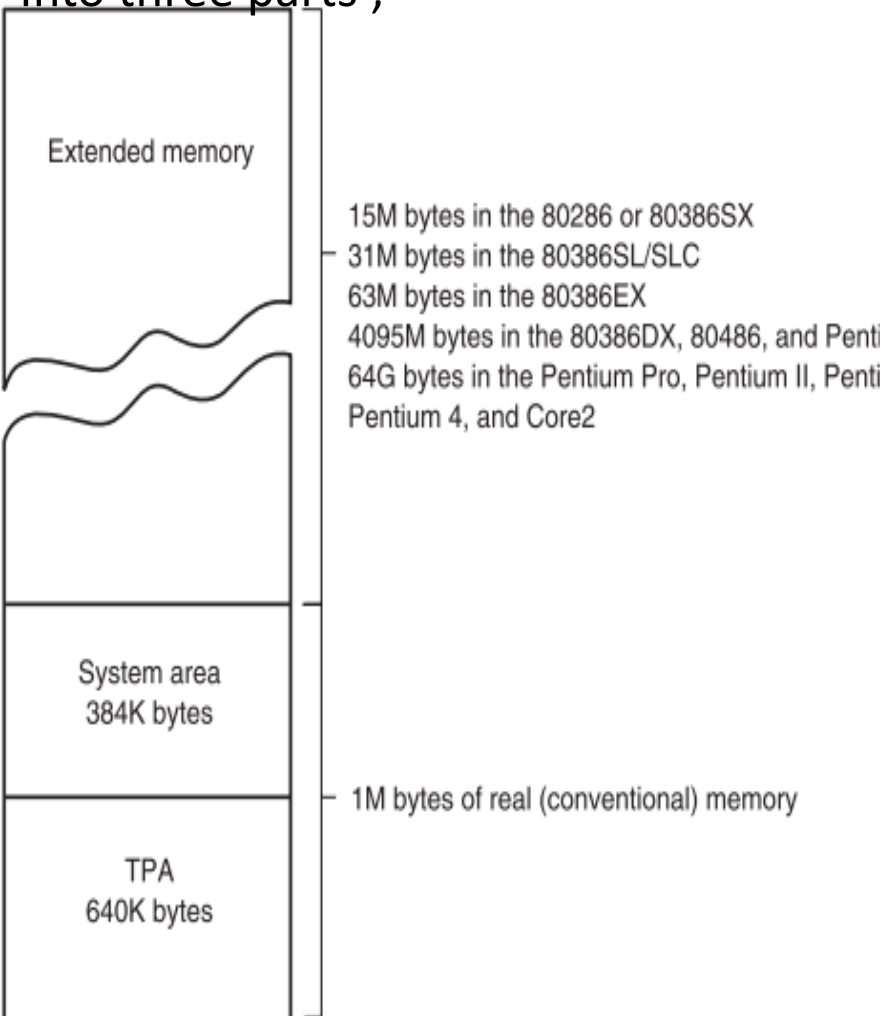


The Memory System

➤ Primary Memory (Main Memory)

➤ Secondary Memory

The main memory system is divided
Into three parts ;



Transient Program Area (TPA) holds the DOS operating system and other programs that control the computer. The TPA also stores any currently active or inactive DOS application programs.

The System Area (DOS) contains programs on either a read only memory (ROM) or flash memory (EEPROM), and areas of read/write (RAM) memory for data storage.

DOS (Disk Operating System) Controls the way that the disk memory is organized and controlled, as well as the function and control of the some of the I/O devices connected to the system.

BIOS (Basic I/O System) The system BIOS is a collection of programs stored in either a read-only memory (ROM) or flash-memory that operates many of the I/O devices connected to computer system.

The **interrupt vectors** access various features of the DOS and BIOS, and applications.

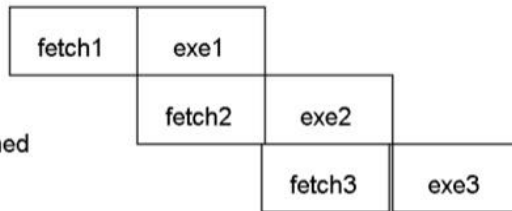
Pipeline

Pipelining in 8086

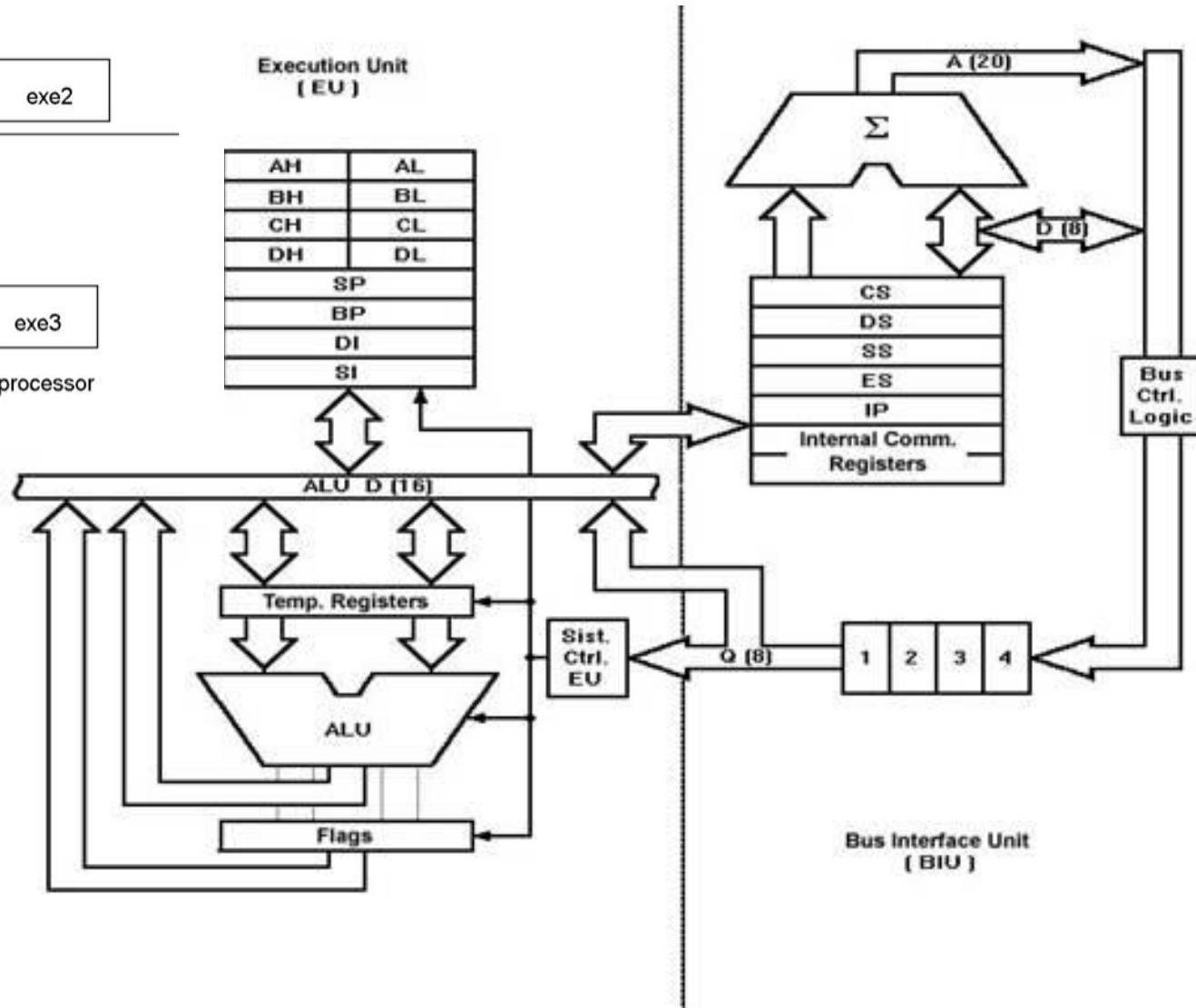
Nonpipelined 8085



Pipelined 8086

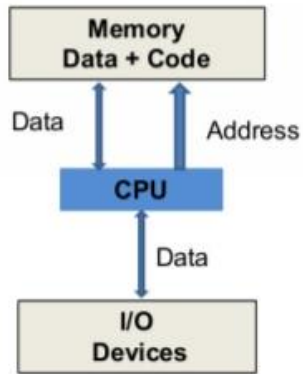


Pipelined in 8086 microprocessor

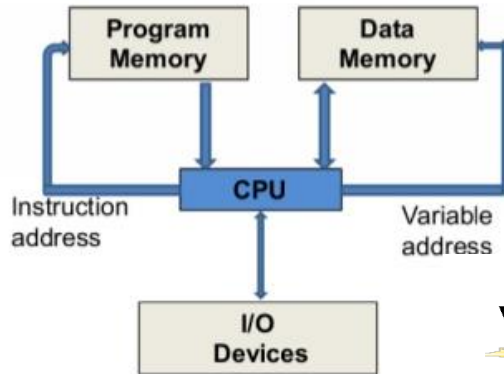


Von Neuman and Harvard Architecture

Von Neumann vs. Harvard Architecture

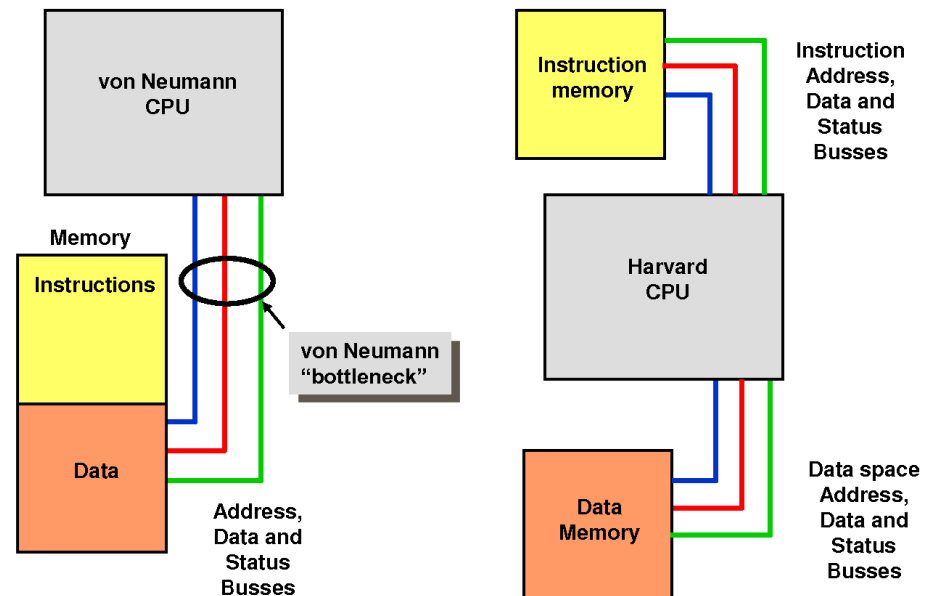


Von Neumann Machine



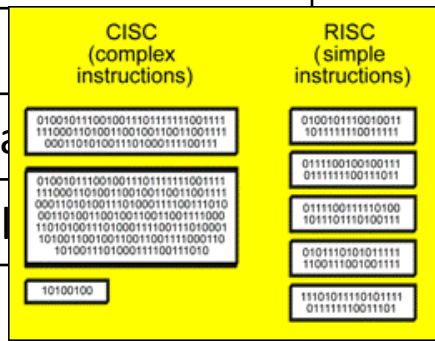
Harvard Machine

von Neumann and Harvard Architectures



CISC and RISC

RISC stand for reduced instruction set computer.	CISC stand for complex instruction set computer.
It has few instructions.	It has more instruction.
Spends more transistors on memory registers .	Transistors used for storing complex instructions.
It has fixed length instructions.	It has variable length instruction.
It has few data types as compared to CISC.	It has more data types as compared to RISC.
It has single cycle operation.	It has multi cycle operation.
Only load/store instruction access memory.	Many instruction access can memory.
It has large number of general purpose registers.	It has small number of general purpose registers and several special purposes registers.
It has few addressing modes.	more addressing modes.
It has a hard wired unit of program	a micro programming unit.
Pipelining does function correctly	ning does not function correctly.



What happens when you switch on a computer



- After switch is pressed
- CPU [registers](#) including [Instruction Pointer](#) (IP) have predefined values.
- In x86 systems the initial value of the [IP is 0xffff0](#) and the instruction stored at this memory location is executed.
- The instruction is [JMP](#) (JUMP) to a Read Only Memory (ROM) which contains the [BIOS](#) and its code starts executing.
- BIOS first does the [Power On Self Tests](#) (**POST**).
 - POST checks the bios
 - then tests the CMOS RAM
 - If there is no problems with this then POST continues to check the CPU, hardware devices such as the Video Card, the secondary storage devices such as the Hard Drive, Floppy Drives, Zip Drive or CD/DVD Drives.
 - If some errors found then an error message is displayed on screen or a number of beeps are heard. These beeps are known as POST beep codes.
- BIOS find the operating system, which is in the CMOS boot sequence.
- Once the target boot device has been selected, the bios will search for the [Master Boot Record](#) (MBR). If it is searching for a hard drive it looks at cylinder 0, head 0, sector 1.
- MBR consists of a special program, the bootstrap loader, and a partition table.
 - The partition table is actually a list of all the partitions in the hard disk and their file system types.
 - The bootstrap loader contains the code to start loading the operating system.

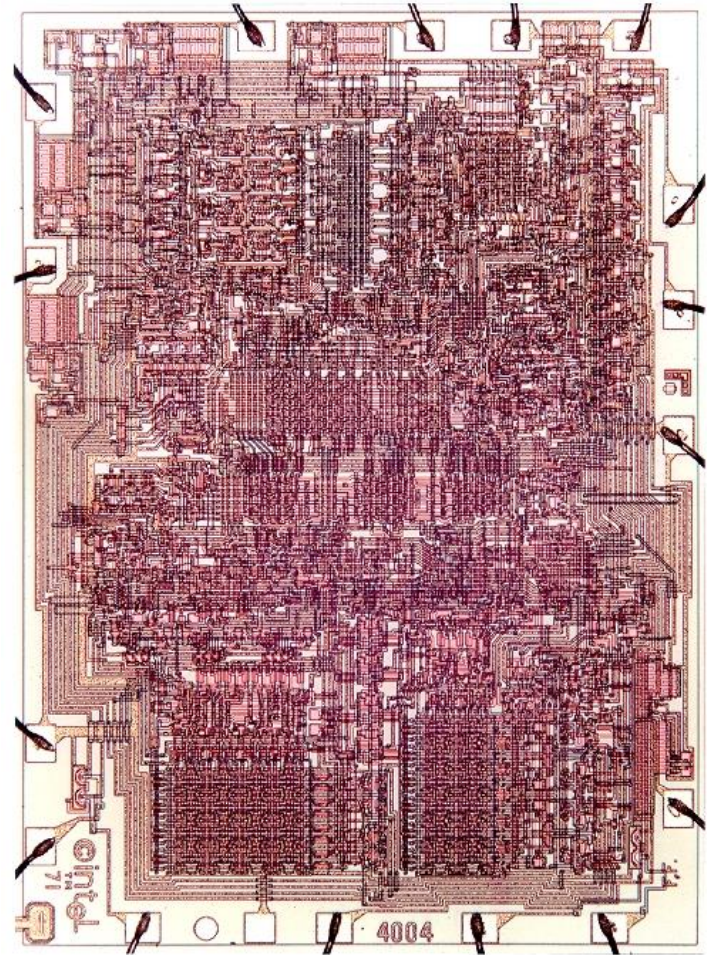
History Overview

- Abacus 500BC
- Mechanical Calculator (Blaise Pascal-1642)
- Mechanical Computing Machinery (Charles Babbage-1823)
Analytical Engine (Steam Powered), Navigational Calculations.
- Electrically driven mechanical calculators (Bomar Brain).
- Herman Hollerith developed punch cards for storing data (1889, also counts, stored and collected information stored on punched cards).
- Konard Zuse, invented first electro-mechanical computer (Z3-1941). It's a relay logic machine clocked at 5.33Hz.
- Electronic Numerical Integrator and Calculator (ENIAC) world first GP programmable electronic computer system (1700 vacuum tubes & 500 miles of wires).

Manufacturer	Part Number
Fairchild	F-8
Intel	8080
MOS Technology	6502
Motorola	MC6800
National Semiconductor	IMP-8
Rockwell International	PPS-8
Zilog	Z-8

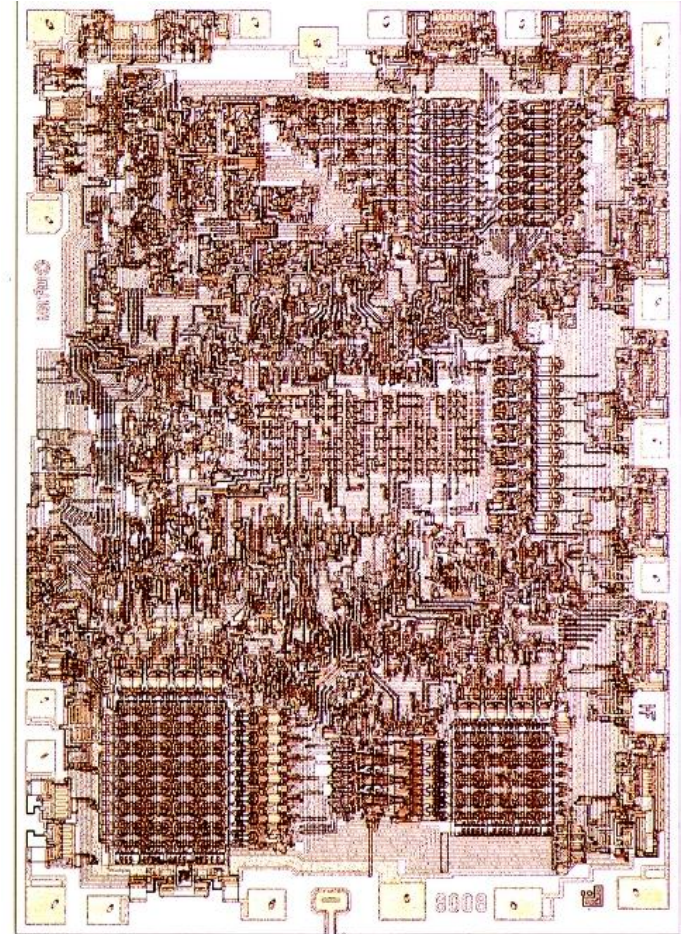
4004

- First microprocessor (1971)
 - For Busicom calculator
- Characteristics
 - 10 μm process
 - 2300 transistors
 - 400 – 800 kHz
 - 4-bit word size
 - 16-pin DIP package
- Masks hand cut from Rubylith
 - Drawn with color pencils
 - 1 metal, 1 poly (jumpers)
 - Diagonal lines (!)



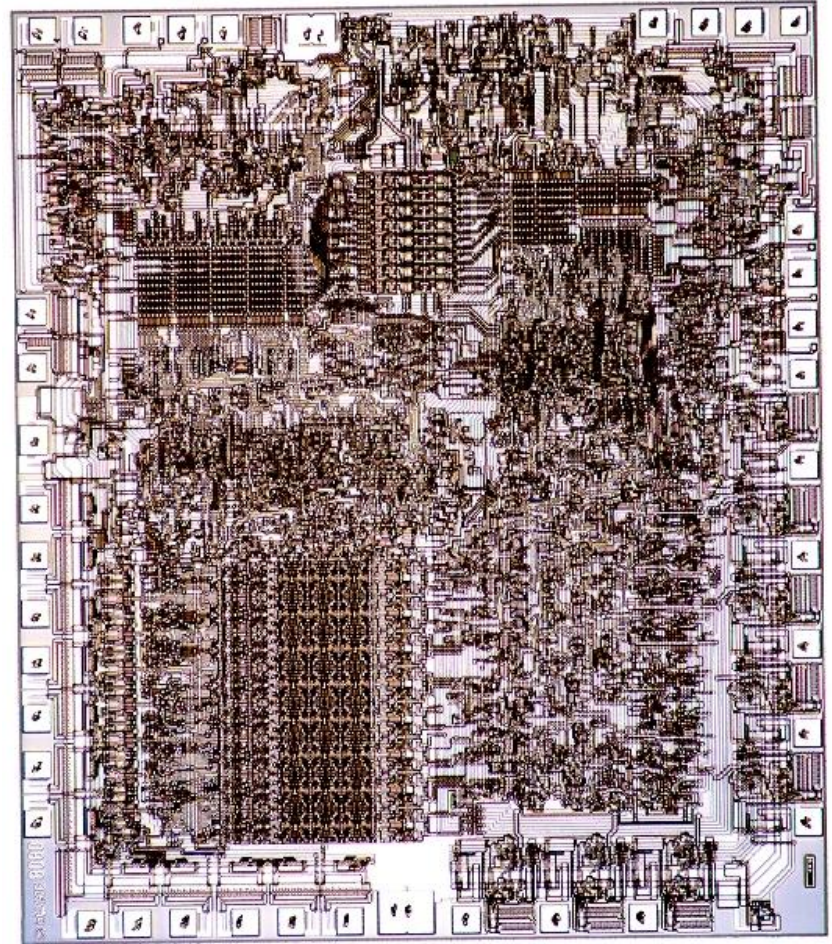
8008

- 8-bit follow-on (1972)
 - Dumb terminals
- Characteristics
 - 10 μm process
 - 3500 transistors
 - 500 – 800 kHz
 - 8-bit word size
 - 18-pin DIP package
- Note 8-bit datapaths
 - Individual transistors visible



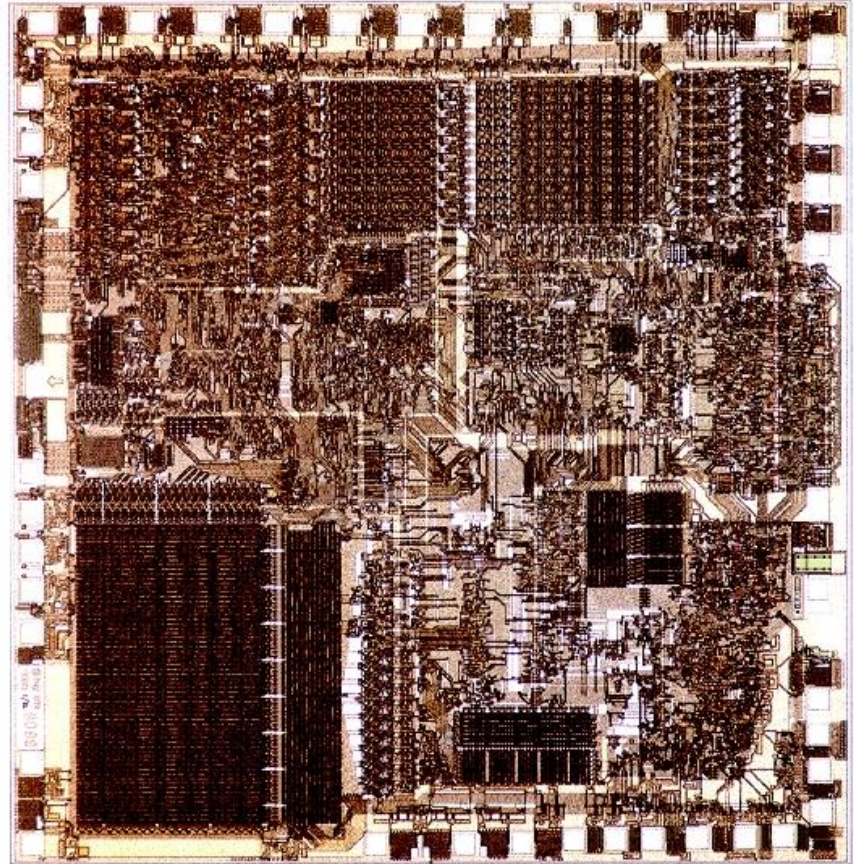
8080

- 16-bit address bus (1974)
 - Used in Altair computer
 - (early hobbyist PC)
- Characteristics
 - 6 μm process
 - 4500 transistors
 - 2 MHz
 - 8-bit word size
 - 40-pin DIP package



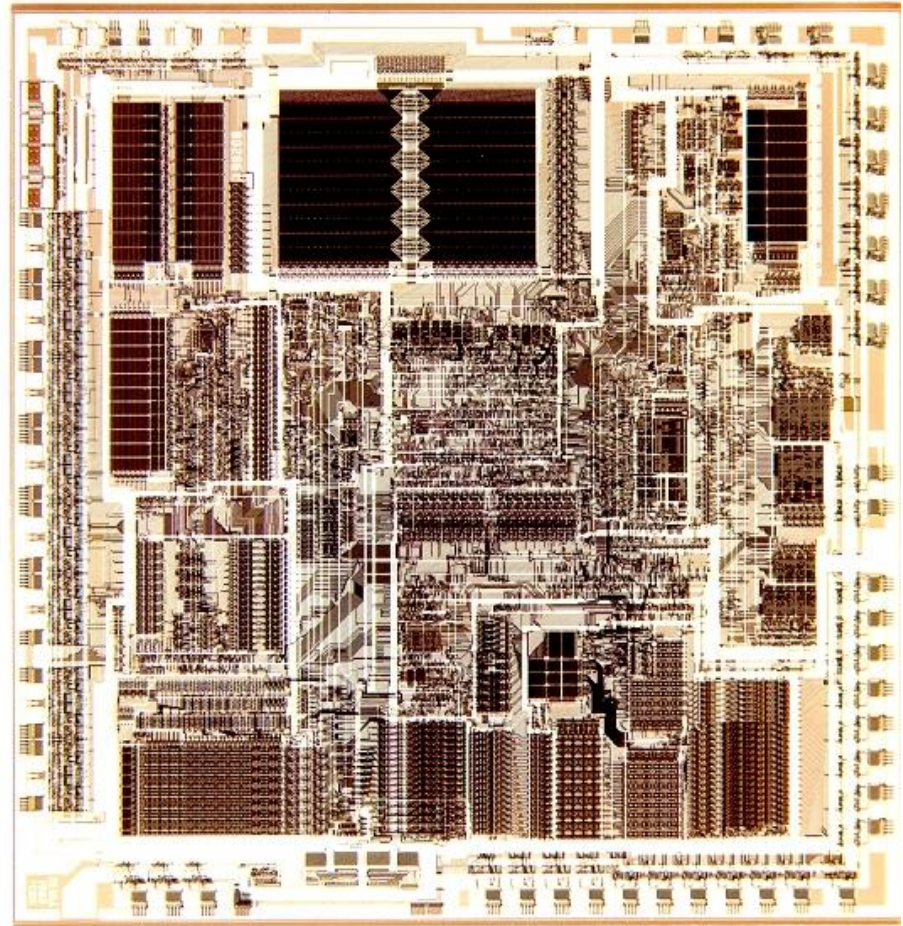
8086 / 8088

- 16-bit processor (1978-9)
 - IBM PC and PC XT
 - Revolutionary products
 - Introduced x86 ISA
- Characteristics
 - 3 μm process
 - 29k transistors
 - 5-10 MHz
 - 16-bit word size
 - 40-pin DIP package
- Microcode ROM



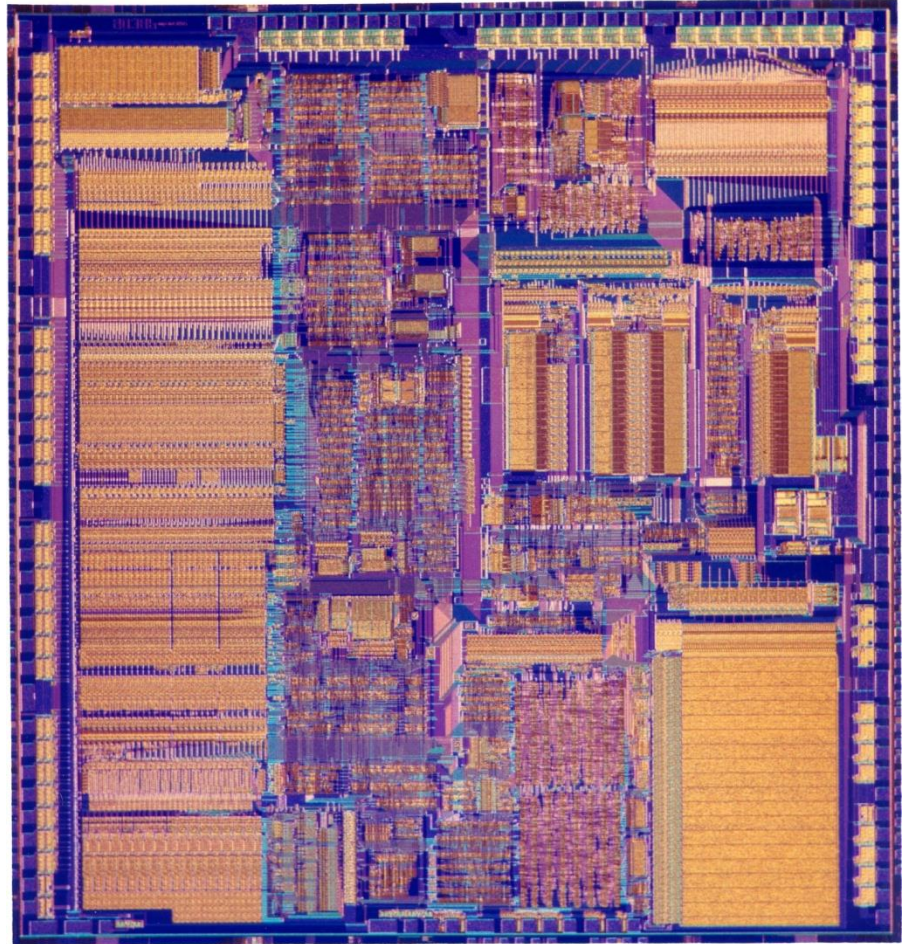
80286

- Virtual memory (1982)
 - IBM PC AT
- Characteristics
 - 1.5 μm process
 - 134k transistors
 - 6-12 MHz
 - 16-bit word size
 - 68-pin PGA
- Regular datapaths and ROMs
 - Bitslices clearly visible



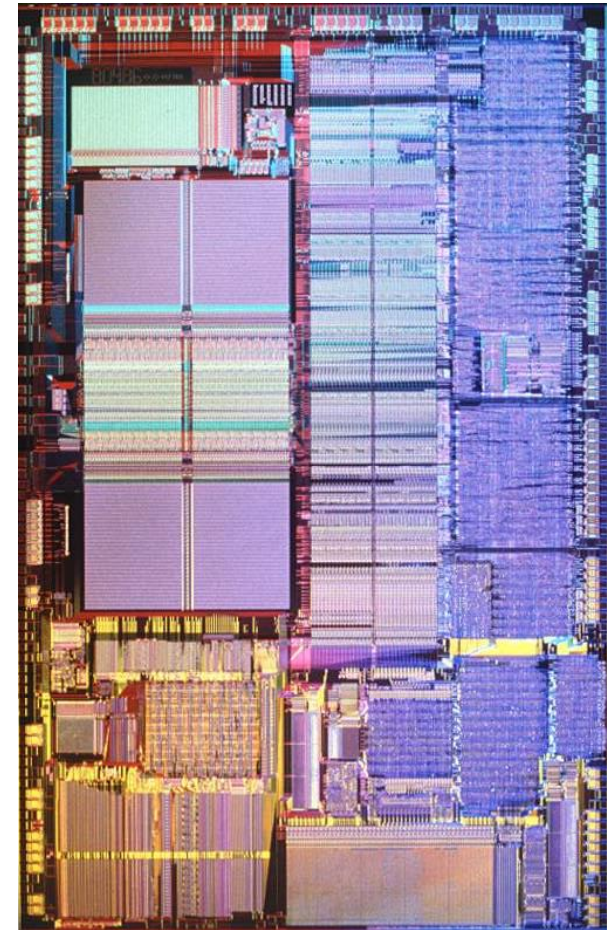
80386

- 32-bit processor (1985)
 - Modern x86 ISA
- Characteristics
 - 1.5-1 μm process
 - 275k transistors
 - 16-33 MHz
 - 32-bit word size
 - 100-pin PGA
- 32-bit datapath,
microcode ROM,
synthesized control



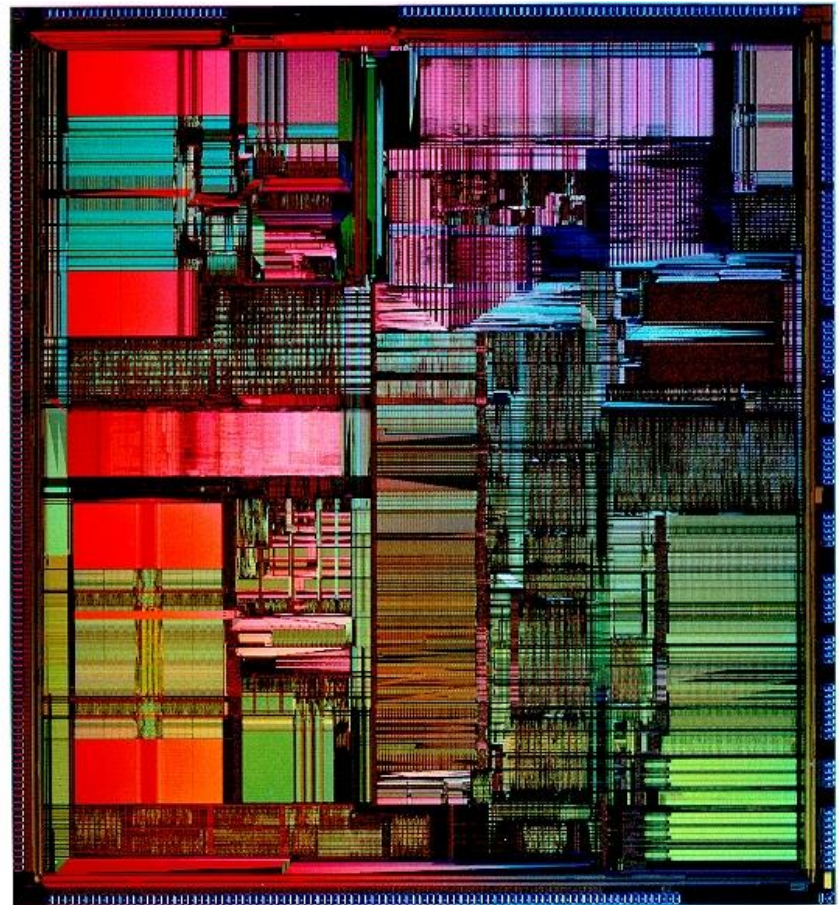
80486

- Pipelining (1989)
 - Floating point unit
 - 8 KB cache
- Characteristics
 - 1-0.6 μm process
 - 1.2M transistors
 - 25-100 MHz
 - 32-bit word size
 - 168-pin PGA
- Cache, Integer datapath, FPU, microcode, synthesized control



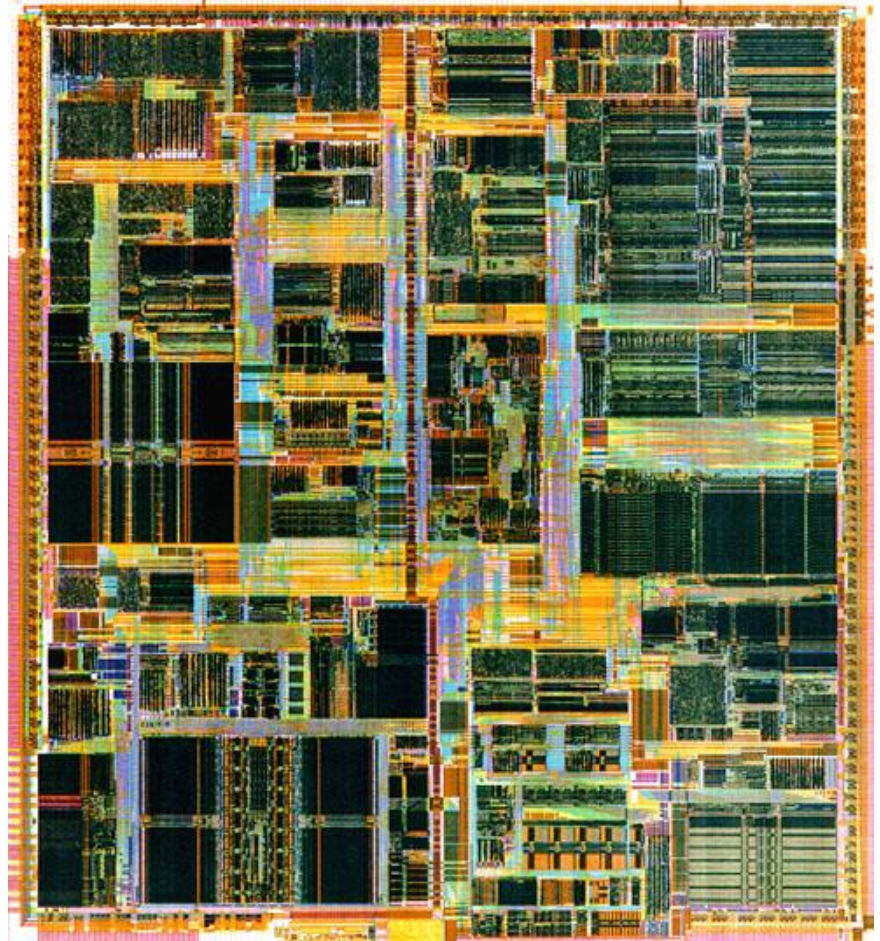
Pentium

- Superscalar (1993)
 - 2 instructions per cycle
 - Separate 8KB I\$ & D\$
- Characteristics
 - 0.8-0.35 μm process
 - 3.2M transistors
 - 60-300 MHz
 - 32-bit word size
 - 296-pin PGA
- Caches, datapath, FPU, control



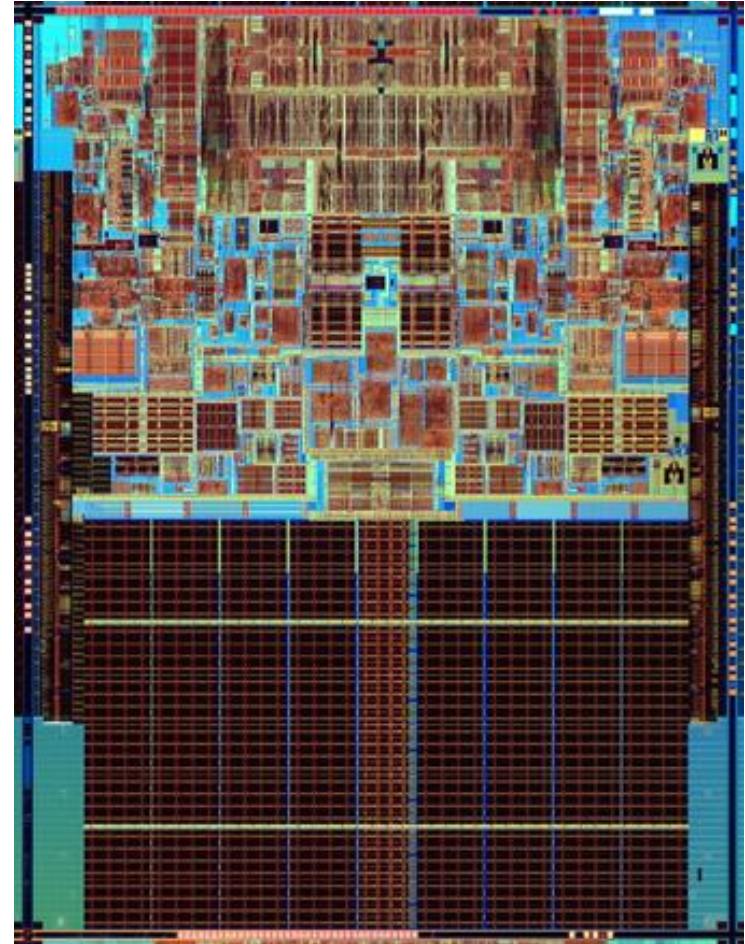
Pentium Pro / II / III

- Dynamic execution (1995-9)
 - 3 micro-ops / cycle
 - Out of order execution
 - 16-32 KB I\$ & D\$
 - Multimedia instructions
 - PIII adds 256+ KB L2\$
- Characteristics
 - 0.6-0.18 μm process
 - 5.5M-28M transistors
 - 166-1000 MHz
 - 32-bit word size
 - MCM / SECC



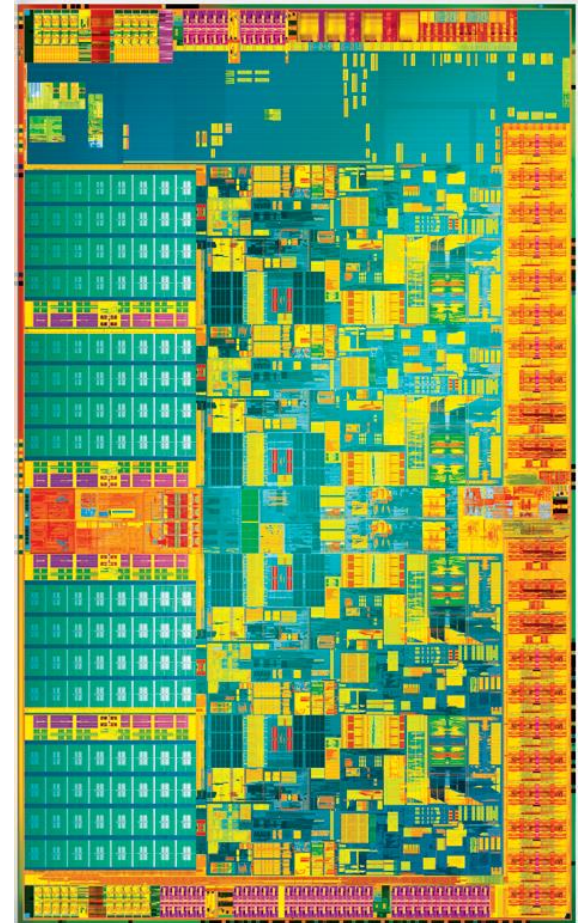
Core2 Duo

- Dual core (2006)
 - 1-2 MB L2\$ / core
- Characteristics
 - 65-45 nm process
 - 291M transistors
 - 1.6-3+ GHz
 - 65 W
 - 32/64 bit word size
 - 775 pin LGA
- Much better performance/power efficiency



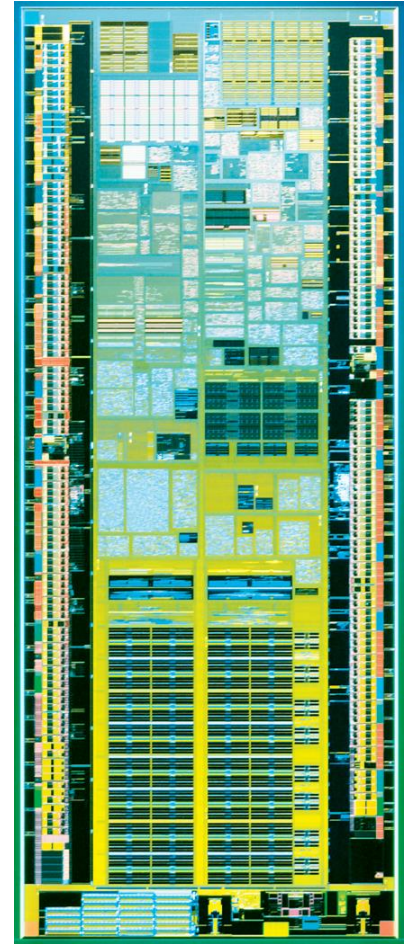
Core i7

- Quad core (& more)
 - Pentium-style architecture
 - 2 MB L3\$ / core
- Characteristics
 - 45-32 nm process
 - 731M transistors
 - 2.66-3.33+ GHz
 - Up to 130 W
 - 32/64 bit word size
 - 1366-pin LGA
 - Multithreading
- On-die memory controller



Atom

- Low power CPU for netbooks
 - Pentium-style architecture
 - 512KB+ L2\$
- Characteristics
 - 45-32 nm process
 - 47M transistors
 - 0.8-1.8+ GHz
 - 1.4-13 W
 - 32/64-bit word size
 - 441-pin FCBGA
- Low voltage (0.7 – 1.1 V) operation
 - Excellent performance/power



Summary (Intel's Processors)

- 10^4 increase in transistor count, clock frequency over 3 decades!

Processor	Year	Feature Size (μm)	Transistors	Frequency (MHz)	Word Size	Power (W)	Cache (L1 / L2 / L3)	Package
4004	1971	10	2.3k	0.75	4	0.5	none	16-pin DIP
8008	1972	10	3.5k	0.5–0.8	8	0.5	none	18-pin DIP
8080	1974	6	6k	2	8	0.5	none	40-pin DIP
8086	1978	3	29k	5–10	16	2	none	40-pin DIP
80286	1982	1.5	134k	6–12	16	3	none	68-pin PGA
Intel386	1985	1.5–1.0	275k	16–25	32	1–1.5	none	100-pin PGA
Intel486	1989	1–0.6	1.2M	25–100	32	0.3–2.5	8K	168-pin PGA
Pentium	1993	0.8–0.35	3.2–4.5M	60–300	32	8–17	16K	296-pin PGA
Pentium Pro	1995	0.6–0.35	5.5M	166–200	32	29–47	16K / 256K+	387-pin MCM PGA
Pentium II	1997	0.35–0.25	7.5M	233–450	32	17–43	32K / 256K+	242-pin SECC
Pentium III	1999	0.25–0.18	9.5–28M	450–1000	32	14–44	32K / 512K	330-pin SECC2
Pentium 4	2000	180–65 nm	42–178M	1400–3800	32/64	21–115	20K+ / 256K+	478-pin PGA
Pentium M	2003	130–90 nm	77–140M	1300–2130	32	5–27	64K / 1M	479-pin FCBGA
Core	2006	65 nm	152M	1000–1860	32	6–31	64K / 2M	479-pin FCBGA
Core 2 Duo	2006	65–45 nm	167–410M	1060–3160	32/64	10–65	64K / 4M+	775-pin LGA
Core i7	2008	45 nm	731M	2660–3330	32/64	45–130	64K / 256K / 8M	1366-pin LGA
Atom	2008	45 nm	47M	800–1860	32/64	1.4–13	56K / 512K+	441-pin FCBGA